

sonar and high-accuracy pressure sensors for better flight.

Drones meant for use by researchers take things a step ahead when it comes to flight systems. AscTec Firefly is designed with that application in mind and therefore comes with advanced sensor components and two ARM7 microprocessors. A low-level processor functions as the data controller and processes all sensor data and sensor-data fusion, while a high-level processor controls the flight system as per the control algorithms given to it. The inertial guidance system drives this drone's autopilot with a 1000Hz update rate.

Hobbypower KK2.15 is an update of the popular HobbyKing KK2.0 flight controller board used by many beginner quadcopter builders. It houses InvenSense MPU-6050 MEMS motion-tracking device designed for low-power use cases.

MPU-6050 combines a 3-axis gyroscope and a 3-axis accelerometer with an onboard digital-motion processor to process 6-axis Motion-Fusion algorithms. The device can access external magnetometers or other sensors through an inter-integrated circuit bus, allowing it access to sensor data without intervention from the system processor.

Can I communicate. Botlink XRD (extended range and data) is an enclosed device that allows you to control your drone through the mobile network instead of using traditional radio or Wi-Fi. This technology allows beyond line-of-sight communication, data transfer and control through your mobile device using mobile telemetry technology that runs on 4G long term evolution (LTE) connections. It also allows serious mission planning by overlaying data of other manned and unmanned aircraft in the area, as well as marking any military or restricted airspace clearly. It also alerts you of adverse weather conditions.

These are especially useful since

Sensors used in drones

3-axis accelerometer. This provides linear acceleration in three axes and plays a very important role in drone stabilisation.

3-axis gyroscope. It provides angular acceleration in three axes and angular motion to the drone.

Magnetometer. This is an electronic magnetic compass that can measure Earth's magnetic field and use it to determine the drone's compass direction (with respect to magnetic north). It is almost always present if the system has GPS input and is available in one to three axes.

Barometer. Since atmospheric pressure changes with the height from sea level, a pressure sensor can be used to give you a pretty accurate reading for the UAV's height. Most flight controllers take input from both, pressure sensor as well as GPS altitude, to calculate more accurate height above sea level.

GPS. It uses signals sent by a number of satellites in orbit around Earth in order to determine their specific geographic locations. UAVs use GPS sensors to autopilot specific route or to automatically return to its take-off point.

Distance. Distance sensors could be used for obstacle sensing and can be ultrasonic, laser or LIDAR based.

—Vikas Kumar Thawani, analog applications engineer, Texas Instruments India, and Naveen Bevara, analog applications engineer, Texas Instruments India

this technology enables drones to fly beyond the line-of-sight of the pilot. "I figured there has to be some kind of software that can give awareness, let a drone know about things like boundaries and weather," says Shawn Muehler, COO of Botlink, to *The New York Times*.

While some drone makers now have dual-control capability that lets two users control a drone—one controls the flight and the other controls the camera—other drone makers like 3D Robotics enable their drones to be smart enough to take over the second role. You can either let the drone know the path that it has to follow while you take over the camera or you can let it lock onto the subject while you play around with the flight to get that perfect angle.

Opening a new frontier for consumers: the sky

Better-engineered drones enabled by the latest technologies could easily be just the right ingredient to unleash creativity on a massive scale. If all these sensor packages

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Parting thought

"But what... is it good for?"

—Engineers in the advanced computing systems division of IBM, 1968, commenting on the microchip

and camera payloads are too boring for you, North Dakota in the USA has legalised drones armed with tasers, rubber bullets or sound canons. The race is now on to build one.

While surveillance and cinema drones are applications that have almost been done to death, there are many new challenges that a drone with the right tools can solve. Something that looks hopeless today could be the next revolutionary application—it all depends on what the engineer in you makes it to be. **EFY**